

Generative Artificial Intelligence and Finite Mathematical Paradigms for Deep Brain Stimulation in Dementia

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Abstract: This paper presents a novel integration of generative artificial intelligence coupled with finite mathematical paradigms to model longitudinal cognitive trajectories in dementia via generalized Deep Brain Stimulation (DBS). We propose an optimal treatment protocol alongside finite element analysis (FEA) of cortical surface manifolds for precise radiofrequency electromagnetic field shaping.

1. Introduction

Dementia phenotypes exhibit nonlinear temporal progression that confounds static Deep Brain Stimulation models. Leveraging finite mathematical decay models paired with Hebbian plasticity constraints, we redefine DBS parameters incorporating generative AI retention limits.

2. Finite Math Equations for Cognitive Trajectory

We model the baseline cognitive severity decay $S(t)$ utilizing an initial state S_0 modulated by a dynamic coefficient β mapped across a 60-month finite interval. The core mathematical formulation operates as:

$$S(t) = S_0 * \exp(-(beta * (1 + 0.1 * \sin(t))) * (t/12)) + \Delta_{DBS}$$

Where Δ_{DBS} is the regenerative Hebbian influence governed by generative parameter amplitude, given by:

$$\Delta_{DBS} = (Amplitude * 0.4) * \Phi_{AI} * (1 - \exp(-t/12))$$

3. Cortical FEA and Toroidal RF Topology

To optimize field specificity and limit isotropic conduction bleed, a finite element analysis of a cortical icosahedron proxy operates with a toroidally bound RF node mesh. The electromagnetic flux (Φ_{RF}) constraint functions as:

$$\Phi_{RF}(r, \theta) = \sigma(r) * B_0 * \cos(\theta + \phi_{var}) + \sum(\epsilon_{EM})$$

Conclusion

Integrating temporal finite regression math via generative constraints with high-fidelity radiofrequency electromagnetic volumetric mapping delivers unprecedented theoretical yield for non-destructive staging of dementia via deep brain stimulation architectures.